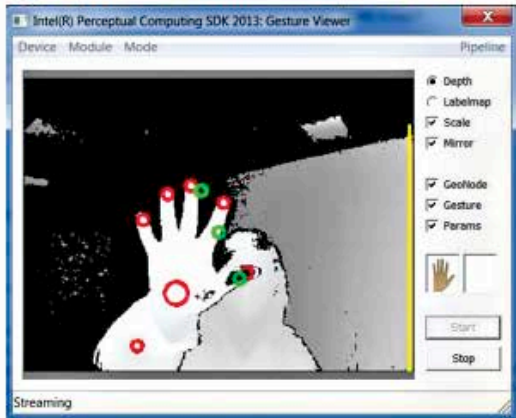


2013:

Fingertips, Palm Center, Grasp Point, Forearm



2014:

Track Additional Points



The new finger tracking module supports 22 points of hand and joint tracking

recognise expressions and emotions like anger, surprise and frustration, on a face that's within the camera's visual range.

HAND AND FINGER TRACKING

The RealSense SDK supports 22 points of hand and joint tracking to support gestures and finger tracking within a range of 0.2m-1.2m. This can be used to provide a very sophisticated means of interacting with the application – including a much more granular GUI control. It enables hand orientation detection and fingertip tracking at the same time, a useful feature that provides control to users for interacting with the application in mid air, as long as their hands are within the specified range. A list of common actions that can be performed with Gesture Recognition is available as examples in the documentation.

GESTURE RECOGNITION

Gesture recognition works with hand and finger tracking to identify dynamic gestures such as grab, zoom, move, swipe, push etc. as a way for you to interact with your application. It is this ability to detect that helps you identify up to 8 static poses and 6 dynamic gestures. They provide an innovative way by which you can gesticulate without any complicated tracking mechanism.

WHAT'S NEW IN 2014

In 2014, 3D Printing and Augmented Reality (AR) have been the talk of the town, and Intel –

EXPERIENCE NEW LEVELS OF INTERACTIVITY AND THE ABILITY TO MANIPULATE BACKGROUNDS AND ENVIRONMENTS IN REAL TIME. THE POSSIBILITIES ARE LIMITLESS WITH REALSENSE

as always – is at the forefront of such technologies working to bring them to users in a cheap and easily available way. The Intel® RealSense™ SDK has been growing at a fast pace and a number of its modules have been developed and integrated in partnership with the best in the field. This ensures that the aim of integrating 3D Printing and AR into our day-to-day devices such as ultrabooks and tablets is achieving its mark, as it's never done before. Some of the new features added this year to the SDK have been mentioned here, although they may not be available in the SDK just yet, but you can stay tuned to receive updates on the same.

AUGMENTED REALITY

In conjunction with Metaio, Intel has developed Augmented Reality features, which can be brought to life in the form of various practical or gaming applications such as Interactive Storytelling. Such features also allow you to create

a 3D virtual world that's aware of the user thus personalising content according to his/her presence. The tracking and recognition engine from Metaio works in combination with the 3D RealSense camera to offer depth data, bringing new levels of interactivity and the ability to manipulate backgrounds and environments in real time. The users could, for example, scan their room and rearrange the furniture in the virtual world.

3D SCANNING AND PRINTING

The new 3D Scanning and 3D Printing technologies in the RealSense SDK have been developed in conjunction with 3D Systems Inc. They offer the possibility of scanning 3D objects into your computer (with the help of a multi-sensor array of cameras) and porting them to the virtual world as 3D models, which can be manipulated as needed. What's more, you can then perform 3D editing and printing of the same or any other 3D model that's been generated by your application. The Intel® RealSense™ SDK enables easier access to these professional quality features, which are now packed into one place.

ENHANCED UNITY SUPPORT

The new RealSense SDK in 2014 completely rethinks the support it offers and has bundled the same for Unity. It adds the Unity Productivity Extension, which provides configurable actions/rules based on the capabilities provided in the SDK. Further, it allows you to write less